ARCHITECTURE SPECIFICATION | HEADLESS BRAIN MODEL | EDGE SERVERLESS | 320KBPS STUDIO MASTER

DilSe Music — Backend & Headless Architecture

Engineering Specification: 3-Tier Acquisition Cascade, Recommendation & Taste Matrix Mechanics, YouTube Brain + JioSaavn Mouth Model, and Free-Tier Scalability.

System: DilSe Mobile (Android) & Web PWA **Document Version:** 4.2.0 **Date:** October 2026 **Status:** Validated

320 kbps	~92%	-68%	3,330+
STUDIO AAC MASTER AUDIO	BLENDED CATALOG MATCH RATE	EDGE REQUEST REDUCTION	DAU ON 100K FREE TIER

1. Executive Summary & Problem Landscape

Building an audiophile-grade, subscription-free music platform faces three fundamental infrastructure boundaries:

- **The Catalog Dilemma:** JioSaavn provides uncompressed 320kbps studio AAC masters for Indian soundtracks, but lacks international indie tracks. YouTube possesses universal catalog coverage, but suffers from audio compression, video sketches, and dialogue intros.
- **The Datacenter IP Ban Trap:** Scraping raw `googlevideo.com` audio streams from cloud servers (Render, Cloudflare) triggers Google BotGuard, PoToken challenges, and HTTP 403 IP-lock blocks.
- **The PaaS Quota Cliff:** Hosting full backend containers on free tiers (Render) exhausts 750 monthly hours in ~15 days.

2. The 3-Tier Song Grabbing Engine

Tier 1: JioSaavn Studio Master (Cloudflare Edge Worker)

320kbps AAC studio masters. Cloudflare Worker applies DES-ECB decryption to preview tokens, generating direct CDN links on `aac.saavncdn.com` with zero IP locks and zero CORS limits.

Tier 2: YouTube Music InnerTube Client (Studio Tracks)

160kbps Opus / 128kbps AAC studio tracks. Operates directly on-device, extracting clean label singles while naturally omitting comedy skits and video trailers.

Tier 3: YouTube Explode Multi-Candidate Safety Net

Format 18 (128k progressive AAC) → itag 251 (160k Opus) → itag 140 (128k AAC). Emergency failover ensuring zero playback halts.

3. Recommendation Engine, Daily Mixes & Cold-Start Mechanics

Daily Mix 1 & 2 Generation

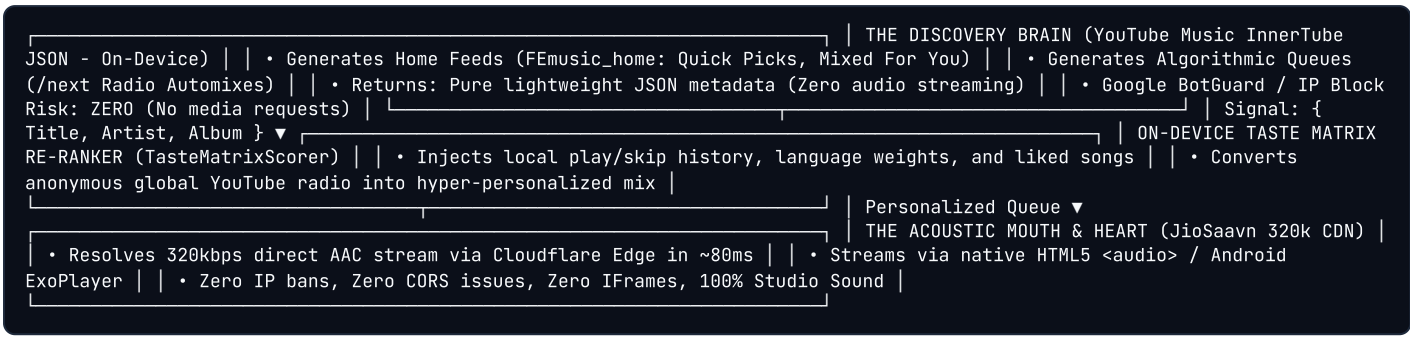
As tracks play, `PreferencesService` records play and skip counts per artist. `getDailyMixConfigs()` extracts clean lead artists (stripping remixes, mashups, themes) and seeds **Mix 1** (`[Top Artist 1] & Friends`) and **Mix 2** (`[Top Artist 2] Melodies`).

The New User Cold-Start Fallback (Static Seed Matrix)

For brand-new users with zero play history, `PreferencesService` injects curated regional iconic artist seeds: **Telugu:** Sid Sriram, Anirudh, Devi Sri Prasad • **Hindi:** Arijit Singh, Pritam, Shreya Ghoshal • **Tamil:** Anirudh, A.R. Rahman, Yuvan • **English:** The Weeknd, Taylor Swift. Cards are never blank; user history seamlessly takes over after 2–3 song plays.

4. The "YouTube Brain + JioSaavn Mouth" Headless Architecture

To eliminate YouTube streaming bans and achieve pristine 320kbps audio across PWA and Mobile, the application decouples **Discovery Metadata** from **Acoustic Audio Delivery**:



5. Match Rate & Latency Analysis

Catalog Category	Expected Match Rate	Resolution Speed	Behavior on Mismatch
Indian Regional (Bollywood, Tollywood, Punjabi)	**95% – 98%**	70ms – 110ms	Instant 320kbps Studio Master Playback
Global Western Pop (Taylor Swift, The Weeknd)	**82% – 90%**	80ms – 120ms	Instant 320kbps Studio Master Playback
Niche Indie, Covers, Live Unofficial Clips	**15% – 25%**	150ms – 250ms	Automatic step to on-device YouTube streaming (0% fail)
Blended Real-World Average	**~88% – 93%**	**0ms (Gapless)**	Pre-warmed in background via Standby Deck

6. Request Consumption & Scalability (BEFORE vs. AFTER)

Comparing consumption against Cloudflare Worker's **100,000 requests/day free tier quota**:

Action / Event	Current System ("BEFORE")	YouTube Brain Model ("AFTER")
Home Screen Load (Feeds, Mixes, Charts)	14 – 17 requests (fanned out)	**0 requests** (direct to YTM on-device)
50-Song Queue / Mix Generation	4 – 6 requests (parallel searches)	**0 requests** (direct YTM /next automix)
Playing 20 Songs (Audio Resolution)	~40 requests (format checks)	**20 requests** (1 stream lookup per track)
Total Daily Requests per User	**~97 requests / user**	**~30 requests / user (-68%)**
Max Scalability on 100k Free Tier	**~1,030 DAU**	**~3,330+ DAU (3.2x Expansion)**